Introduction

The Beach-fx modeling system requires application of a one-dimensional coastal morphology change model that is suitably accurate for predicting beach profile change under storm attack. The original framework made use of the SBEACH model, an empirical representation of morphodynamics introduced in Larson and Kraus (1989). The option to use the CSHORE model was outlined in Johnson and Sanderson (2020) and the scripting code is provided
https://github.com/erdc/BeachFX_CSHORE_link
Both models require comparable bottom position initial conditions with wave and water level seaward boundary data. In addition to profile response, SBEACH and CSHORE also provide hydrodynamic and wave estimates required by the Beach-fx model. While the models are generally interchangeable, they are contrasting in foundational principles and associated limiting assumptions. A short review of the operational model differences is provided herein to assist in rational model choice.

Model Formulation

Both SBEACH and CSHORE operate under the assumption of longshore uniformity, and wave and current estimates derive from the numerical integration of the depth-averaged energy, momentum, and continuity equations.

The predictions of sediment transport and associated morphology change are fundamentally different, however. The SBEACH model pre-determines an empirically-based equilibrium profile on the basis of sediment, wave and water level conditions. The rate of bottom position evolution, approaching the prescribed steady-state, is based on the degree of departure from equilibrium and the hydrodynamic forcing. With a basis in prescribed steady profiles, The SBEACH model lacks foundational rigor but is inherently robust and stable.

The CSHORE model, conversely, posits process-based sediment transport algorithms for a nearshore breaking wave environment. The model provides distinct estimates for bedload, suspended load, and wave-related sediment transport. Generally speaking, when applied in an appropriate manner, the CSHORE model provides more accurate results [e.g. Johnson etal 2012]. The underlying assumptions for the CSHORE model are, however, more restrictive, and predictions resulting from conditions in violation will likely be in gross error. A partial accounting of model assumptions to be regarded is:

- Sediment grain size varies in the range $.1mm < d_{50} < 1mm$,
- Minimal bed position change at the seaward boundary, equivalent to having a seaward boundary sufficiently deep to avoid breaking,
- Dune overtopping is limited to intermittent wave overtopping. This precludes having conditions where the seaward boundary water level is higher than the maximum profile position.
- Slopes are limited to the sediment angle of repose. This precludes the use of vertical walls, which are not currently supported within CSHORE.

Model Operation

At present, the SBEACH workflow is designed to execute on a Windows platform in serial. It is common practice to manually subdivide a long series of Monte Carlo simulations and perform the morphology computations on separate individual computers and manually recombine the results at completion. In contrast, the CSHORE/BEACHFX implementation is scripted for rudimentary parallel execution, where an arbitrary number of discreet processors or cores can be used simultaneously on a desktop computer or HPC resources. While the HPC platform can pose challenges, the parallel system affords considerable efficiency.

References

Johnson, B.D. Sanderson, D.R. (2020) *On the Use of CSHORE for Beach-fx*, US Army Corps of Engineers Technical Note CHETN-II-59

Johnson, B.D., Kobayashi, N., and Gravens, M.B. (2012) *Cross-Shore Numerical Model CSHORE for Waves, Currents, Sediment Transport and Beach Profile Evolution*, US Army Corps of Engineers Technical Report TR-12-22.

Larson, M., Kraus, N.C. (1989) *SBEACH : numerical model for simulating storm-induced beach change. Report 1, Empirical foundation and model development* US Army Corps of Engineers Technical Report; CERC-89-9 Report 1